

Measuring Validator and Sandboxed-Execution Coverage for LLM-Generated NetOps Artifacts: A Paired Confirmatory Study

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CNSM 2026 Agentic AI Researcher Track

Abstract—We preregistered and executed a paired confirmatory experiment (study_id f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e) that measured the marginal detection benefit of adding sandboxed emulation to a deterministic validator for LLM-generated intent→configuration artifacts. Under shared_initial_candidate semantics using the declared model "gpt-5-mini" (provider: openai_responses) we evaluated Npairs = 720 confirmatory tasks. The deterministic validator (deterministic_netops_validator_v5) flagged 719/720 = 0.9986111111111111; the guarded pipeline (validator + sandboxed emulation) flagged 720/720 = 1.0. Paired difference (guarded - baseline) = 0.001388888888888884; exact McNemar two-sided p = 1.0; 95% bootstrap percentile CI (10000 resamples, seed = 1729) = [0.0, 0.004166666666666667]. Run accounting: model_calls_used = 721 (720 shared initial + 1 repair-stage call). These artifact-grounded results lie far below our preregistered minimum effect-of-interest (0.20); we report the preregistered design, exact quantitative outcomes, run-accounting diagnostics, artifact-grounded interpretation for the negligible guarded benefit in this regime, and reproducibility pointers into the archived execution bundle.

I. INTRODUCTION

Generative large language models (LLMs) are increasingly proposed to translate operator intent into device configuration and NetOps workflows. Operational deployment requires generated artifacts to satisfy syntactic correctness, workflow ordering, and higher-order safety/policy constraints. A common mitigation architecture is generate→validate→(repair)→execute, sometimes augmented by sandboxed emulation to reveal runtime-only failures that static validators miss. Prior surveys and system reports advocate such workflow-centred mitigations [1]–[3], but provide limited large-scale paired measures of the marginal value added by sandboxed execution. Motivated by this gap, we preregistered and executed a controlled paired study to measure the aggregate detection-rate difference contributed uniquely by an automated sandboxed-emulation stage when the same initial generated candidate is analyzed by both conditions.

Research question and contributions. Our preregistered research question was: for synthetic intent→configuration tasks under shared_initial_candidate semantics, what is the paired detection-rate difference (guarded - baseline) between (A) a deterministic validator-only pipeline and (B) the same deterministic validator followed by automated sandboxed emulation? Contributions: (1) a machine-readable preregistration and faithful autonomous execution; (2) an artifact-grounded

paired analysis on Npairs = 720 with complete accounting; (3) an artifact-grounded interpretation explaining a near-zero marginal effect; (4) concise reproducibility pointers into the archived bundle and prescriptive boundary conditions where guarded stages are likely to matter.

II. RELATED WORK

Workflow-centred mitigations and verification tools. Recent surveys and architecting reports document a common pattern for LLM-enabled NetOps: pair generation with deterministic validators, formal verifiers, or emulation to reduce unsafe or incorrect device-level actions [1], [2]. Tool- and system-focused work demonstrates concrete instantiations: configuration-analysis engines (e.g., Batfish-style approaches and NetKAT-based verifiers) provide rule-driven semantic checks over device configurations and have been extended in prototypes that combine generation with symbolic or executable verification [2], [3]. Parallel lines of work emphasize the role of sandboxed execution (Mininet/Mininet+GNS3 hybrids, lightweight emulators) to surface dynamic, stateful failures that static validators cannot express, particularly for make-before-break or multi-step state transitions [4], [5].

Coverage heterogeneity and verifier ceilings. Prior empirical and methodological studies highlight that validator gains depend strongly on the validator rule-set, the task abstraction, and the emulator fidelity. For instance, verification-oriented projects show that many common configuration errors (syntactic mistakes, missing required commands, simple ordering violations) are well-covered by high-quality static validators, whereas subtle runtime invariants or environment-dependent failures (timing-dependent race conditions, stateful interaction errors) often escape static analyses and require emulation or more expressive specification languages [2], [6]. This heterogeneity creates two operationally important regimes: (i) validator-dominated regimes where a broad, rule-rich validator attains a high true-positive rate and leaves little headroom for emulation to add detections, and (ii) runtime-detection regimes where emulators or high-fidelity execution are necessary to reveal violations that are intrinsically dynamic.

LLM interaction patterns, mistake localization, and repair. Empirical studies in adjacent program- and specification-generation tasks show a recurring pattern: LLMs may fail to locate their own reasoning errors reliably but can perform targeted corrections when errors or locations are externally

signalled or when a verifier pinpoints a violation [5], [7]. This motivates generate→validate→repair pipelines that use validators or emulators as external mistake detectors and—optionally—invoke constrained repair-stage model calls. The present study enforces `shared_initial_candidate` semantics to isolate exactly these downstream contributions (validator vs. emulator vs. explicitly recorded repair calls) and avoid conflating generation variability with guarded-stage effects.

Benchmarks, evaluation gaps, and operational implications. Several surveys and position papers argue that existing LLM benchmarks insufficiently capture workflow-centred safety properties (rollback-aware scoring, multi-device transactional integrity, emulation-confirmed invariants) and call for niche benchmarks that combine intent→generate, static verification, and sandboxed execution to reflect operational trustworthiness [1], [6]. Our confirmatory paired design directly addresses this suggested evaluation axis by quantifying the marginal detection contribution of an emulation stage under tightly controlled shared-candidate semantics; the negligible aggregate gain we observed is consistent with a validator-ceiling interpretation in a curated synthetic task bank and illustrates the evaluation-risk of reporting single-stage improvements without workflow-centred controls.

III. METHODOLOGY

Preregistration and confirmatory design. We preregistered a paired-binary confirmatory design (study_id f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e). The primary estimand is the `paired_success_rate_difference_guarded_minus_baseline` aggregated across confirmatory samples. The preregistered sample plan targeted $N_{\text{pairs}} = 720$ (planned episodes = 1440) with a minimum effect-of-interest of 0.20 (20 percentage points) for power planning. The preregistration fixed inclusion/exclusion rules, missingness handling (deterministic reseed up to two times for parser failures; deterministic replacement for irrecoverable failures), multiplicity handling for exploratory secondary tests, and contamination controls based on deterministic hashing and artifact-version checks.

Task generation, topology templates, and vendor abstractions. Confirmatory items were generated deterministically from the `intent_configuration_repair_v1` task family using a deterministic task generator. The bank covers three abstracted vendor-CLI families and three topology templates (leaf–spine, 3-node service chain, triangle). Each task manifest entry contains: `initial_state`, `expected_final_state`, an explicit intent, a `required_command_sequence`, `allowed_touched_fields`, and workflow policies (e.g., `make-before-break`, `must-be-down-for-fields`, `minimum_active_in_groups`). Inclusion required vendor-specific parser acceptance after up to two deterministic reseeds. In the confirmatory run there were zero irrecoverable parser exclusions and `complete_pair_count = 720`, consistent with the preregistered missingness policy.

Model, sampling semantics, and `shared_initial_candidate` enforcement. The declared/requested model was recorded as "gpt-5-mini" (provider: `openai_responses`) and the model configuration archived (requested `max_output_tokens = 2000`;

the client did not set an explicit temperature parameter—recorded as null in the model configuration). Important pre-registered clarifications: (a) temperature recorded as null indicates the client supplied no explicit temperature value; it does not imply provider-side determinism or `temperature=0`, and it does not alter the `shared_initial_candidate` contract; (b) `shared_initial_candidate` semantics were enforced per the preregistration: for each pair we obtained exactly one sampled initial candidate (a single provider call) and that identical candidate was presented to both downstream conditions (baseline and guarded). This design isolates downstream guarded-stage effects by removing generation variability as a source of between-condition discordance.

Baseline validator specification. The baseline validator (`deterministic_netops_validator_v5`) implements deterministic, rule-based intent-constraint validation. Its core checks (applied to a normalized candidate) include: vendor-CLI syntactic parsing; per-command parsing and canonicalization; sequence-order checks against `required_command_sequence`; atomicity/workflow constraints (e.g., `make-before-break`, `all_down_before_any_field_change`); protected-interface invariants; group-level activity constraints (minimum/maximum active interfaces within specified groups); and final-state conformance tests (`expected_final_state`). The validator returns a binary detection flag (detected/not-detected) under the preregistered scoring rule `full_intent_constraint_validation` and emits a `validator_trace` object (`validation_before/after`, `parsed_command_count`, `required_command_count`, `violation_codes`).

Guarded pipeline and sandboxed-emulation harness. The guarded pipeline runs the baseline validator and, irrespective of the validator outcome, then executes the candidate in an automated sandboxed-emulation harness that instantiates the declared topology template and executes normalized CLI commands against vendor-abstracted virtual devices. The emulation harness records runtime observations relevant to operational safety: interface admin transitions and timing, transient downtime and concurrent-interface states, MTU/VLAN application effects, and observable connectivity or reachability consequences at the topology level. Emulation results are translated into a set of runtime-observation rules that can trigger a guarded detection if they reveal policy or safety violations not encoded in the static validator rule set. The emulation harness is intentionally conservative in mapping observations to detections; its purpose in this study is to reveal runtime-only misconfigurations that the baseline validator might miss.

Repair policy and run accounting. The preregistered repair policy allowed at most one repair-stage model invocation per task; repair invocations are permitted only after the shared initial candidate is generated and after baseline validator and emulation observations are available for the guarded condition. All repair calls and their transcripts are recorded in the execution manifest and deterministic reconciliation artifacts. Provider-call auditing for the confirmatory execution records `model_calls_used = 721` with `stage_counts` showing `shared_initial_generation = 720` and `repair = 1`; thus only

a single repair-stage call occurred across the entire confirmatory set. Because `shared_initial_candidate` semantics were enforced, any discordant pair (guarded-only detection) must be attributable to either an emulation-observed runtime violation or to the single documented repair invocation; both are traceable in the archived per-episode artifacts.

Inclusion, determinism, and contamination controls. The execution used deterministic artifact hashing, generator-seed recording, and explicit binary/configuration-version checks before confirmatory runs. Parser-failure handling used deterministic reseeding with stored PRNG seeds under the task manifest; irrecoverable failures would be replaced deterministically and recorded. Deterministic reconciliation verifies `complete_pair_count = 720`, `completed_episode_count = 1440`, and `failed_episode_count = 0`. Provider-call auditing reports `cache_miss_count = 721` and zero failed hosted model calls. The preregistered contamination rules also halted confirmatory execution if validator or emulator binary/version mismatches were detected; no such mismatches occurred.

Analysis procedures and pre-registered inferential methods. The primary analysis executor is a paired-binary procedure aligned with McNemar-style inference and cluster-robust variance estimation. For the primary aggregated estimand we compute the paired detection-rate difference (guarded - baseline) with a two-sided exact McNemar test and produce a 95% paired-bootstrap percentile confidence interval (bootstrap resamples = 10000, bootstrap_seed = 1729) as specified in the preregistration. Secondary and exploratory per-class/vendor paired analyses were pre-specified and treated with Bonferroni-adjusted confidence intervals in the preregistered multiplicity plan. The primary analysis used the preregistered complete-pair handling (exclude irrecoverable technical pairs) and a pre-registered sensitivity analysis treating irrecoverable failures as nondetections; in the confirmatory execution no irrecoverable pairs were present, so conclusions rest on complete paired data.

Artifact-locator guidance for independent verification. All per-episode raw scoring and trace artifacts were archived. The canonical input-results file is `execution/raw_results.jsonl` (recorded in the execution manifest). Independent verifiers can locate any discordant pair by searching that file for an entry where `baseline_score = 0` and `guarded_score = 1`; `deterministic_reconciliation.json` in the analysis artifacts links pair accounting and `provider_call_audit` entries and identifies the repair-stage call(s) associated with discordant outcomes. Summary paired counts and contingency outputs are archived as `analysis/paired_contingency_table.csv` and `analysis/condition_summary.csv`. The analysis used these archived artifacts as the direct input to the paired-binary executor; the numeric outputs reproduced in Results are verbatim from `analysis/results.json`.

Design rationale and trade-offs. We chose `shared_initial_candidate` semantics to produce a conservative, direct estimate of guarded-stage marginal contributions: by reusing the same sampled initial candidate for both conditions, any observed discordance can only arise downstream of

generation (emulation observation or repair). This isolates the guarded-stage value but intentionally omits any potential benefit that could accrue from independent re-generation in the guarded condition (e.g., prompting-based repair prior to validator execution). Our inclusion policy (parser-validated synthetic tasks) reduces noise and ensures task executability but also concentrates the task bank where a strong validator can achieve high coverage; this design choice is central to interpreting the near-zero aggregate effect observed in the confirmatory run.

IV. RESULTS

Primary confirmatory counts and test statistics (artifact-grounded). Analysis artifacts report the paired contingency counts: `n11 = 719` (detected by both), `n10 = 1` (guarded-only), `n01 = 0` (baseline-only), `n00 = 0` (neither). Marginal detection rates are `baseline = 719/720 = 0.9986111111111111` and `guarded = 720/720 = 1.0`. The paired estimate (guarded - baseline) = `0.0013888888888888884`. Exact McNemar two-sided test `p = 1.0`. The paired-bootstrap percentile 95% CI (10000 resamples, bootstrap_seed = 1729) = `[0.0, 0.004166666666666667]`. These numbers are reproduced verbatim from `analysis/condition_summary.csv` (sha256 394bc690671b731194f9b42b4dcbe28fded6e2ec8cf898b79831225343c22c28) and `analysis/paired_contingency_table.csv` (sha256 efe6e8e7016fa843a8226e911dbde829e943903d719101da8d956b3ff899133c).

Canonical artifact pointers and the unique discordant episode. The complete per-episode raw results are archived in `execution/raw_results.jsonl` (input-results SHA-256 = `df367ecc40a6f0eaf402129260e11d8450884a4ea5fdb94a602e0d2513aef0e8`). The single discordant pair (the unique `n10` item) is the lone entry in `execution/raw_results.jsonl` whose recorded `baseline_score == 0` and `guarded_score == 1`. That `raw_results` entry contains the canonical `pair_id` field and includes the `validator_trace` and `repair_provider_trace` fields that link the guarded detection to a downstream repair-stage provider trace. Deterministic reconciliation and provider-call accounting are archived in `analysis/deterministic_reconciliation.json` (sha256 49731a2e52e049ccf5b53c35ddac2351ad75e7e95504dc4a42bcb096d0dec3aa) and confirm `model_calls_used = 721` and `stage_counts = {"shared_initial_generation": 720, "repair": 1}`. In short: the discordant episode is uniquely identifiable within the archived bundle (`execution/raw_results.jsonl`, sha256 `df367ecc...`) and the same bundle contains the `deterministic_reconciliation` artifact (sha256 `49731a2e5...`) that documents the single repair-stage call that is connected to that `raw_results` entry via its `repair_provider_trace` field.

Run-accounting diagnostics and sensitivity checks. Provider-call audit (`deterministic_reconciliation.json`) reports `hosted_model_call_event_count = 721`, `completed_hosted_model_call_event_count = 721`, `failed_hosted_model_call_event_count = 0`, `cache_hit_count = 0`, `cache_miss_count = 721`. The preregistered

missingness policy and sensitivity analyses were applied; no irrecoverable parser or emulation failures occurred (analysis/missingness_summary.csv documents complete_pairs = 720 and zeros for all failure categories). The primary analysis used the preregistered complete_pair_primary treatment; the preplanned sensitivity (treating irrecoverable technical failures as nondetections) is moot here because there were no such failures in the confirmatory run.

Representative per-episode diagnostics. Representative scoring traces (execution/scoring/task-000001-*.json ... task-000006-*.json) show parsed_command_count values of 4, 2, 6, 4, 3, and 6 respectively and validator_violation_count = 0 for these examples; the validator_version reported in these traces is "5.0". These representative artifacts illustrate that the shared_initial candidates in the confirmatory bank were typically complete with respect to the required command sequences and that validator checks reported validity for the majority of cases. The representative traces are archived in the execution bundle and are consistent with the high baseline success rate.

Attribution of the single discordant detection. Because shared_initial_candidate semantics were enforced (one shared sampled initial candidate per pair), the only proximate causes for the single n10 observation are (A) an emulation observation not encoded as a validator rule that produced a guarded-stage detection, or (B) the single recorded repair-stage invocation documented in provider-call accounting. The raw_results entry for the discordant pair includes the repair_provider_trace and validator_trace fields permitting an auditor to follow the exact artifact-level linkage from the guarded detection to the recorded repair-stage call; both the raw_results file (sha256 df367ecc...) and the reconciliation artifact (sha256 49731a2e5...) are present in the canonical execution bundle.

Implications of the quantitative diagnostics. The observed discordant count (n10 = 1) yields an aggregate paired difference that is two orders of magnitude below our preregistered minimum effect-of-interest (0.20). The bootstrap CI [0.0, 0.004166666666666667] quantifies the narrowness of the realized uncertainty in this curated regime. Combined with run-accounting (model_calls_used = 721; repair = 1) and the representative per-episode diagnostics, the archived evidence supports a ceiling-effect interpretation: the deterministic validator covered the task bank's explicit syntactic and workflow constraints so thoroughly that the guarded-stage emulation had almost no additional aggregate detections to surface.

V. DISCUSSION

Why was the guarded-stage aggregate benefit negligible? The archived evidence supports three artifact-grounded factors.

1) Validator ceiling effect. Deterministic_netops_validator_v5 covered the syntactic and explicit semantic properties encoded in the synthetic task bank (shutdown-configure-restore, make-before-break uplink switchover, paired MTU changes). Baseline success rate ≈ 0.9986 indicates near-universal validator coverage for these

curated patterns, leaving negligible headroom for the guarded stage.

2) Task curation and inclusion policy. Confirmatory items were parser-validated synthetic tasks specifically designed to exercise workflow policies but not constructed adversarially to evade validator checks. This curation concentrates cases inside the validator's coverage envelope and reduces the prevalence of subtle runtime-only failure modes that emulation might reveal.

3) Shared_initial_candidate semantics and limited repair activity. Enforcing one sampled initial candidate per pair isolates guarded-stage contributions but removes between-condition generation variability. Provider_call_audit and deterministic_reconciliation.json record one repair-stage invocation aligned with the single n10 discordant count; thus the observed discordance is attributable to downstream guarded-stage activity recorded in run accounting rather than generation differences.

Operational implications and boundary conditions. For environments where a high-coverage static validator exists and inputs match its expected patterns (parser-validated, constrained intents), adding automated sandboxed emulation may provide negligible aggregate additional detection at scale. This conditional conclusion must not be generalized: in operational regimes with noisier natural-language intents, adversarial inputs, broader vendor diversity, stateful multi-step interactions, or validator gaps, guarded runtime checks and higher-fidelity emulation are likely to provide substantive benefit. The archived per-episode traces support targeted follow-up hypotheses: adversarial task injection, broader vendor-CLI coverage, and complementary validator families should increase the guarded-stage marginal contribution.

VI. LIMITATIONS

- Synthetic task bank and deterministic task generator produce limited ecological diversity and create a validator-ceiling effect; results may not generalize to noisier, adversarial, or production distributions.
- Single declared/requested model family (gpt-5-mini) and single validator implementation (deterministic_netops_validator_v5) limit generalizability across models and validator families.
- Preregistered shared_initial_candidate semantics (one initial generation reused across paired conditions) isolate guarded-stage contribution but remove between-condition generation variability; this design choice constrains discordance sources to downstream guarded-stage observations or repair calls.
- Sandboxed-emulation fidelity relative to vendor/OS production behaviors and large-scale stateful interactions is limited by the emulation harness used; higher-fidelity emulators could change outcomes in other regimes.
- Study power planning targeted a minimum detectable paired difference of 0.20; the observed aggregate effect (~ 0.0014) lies far below that design sensitivity and should be interpreted as a narrowly scoped near-zero finding for

this experimental regime rather than a general negative result for all NetOps workflows.

- Provider-side nondeterminism and hidden caching remain residual uncertainties despite deterministic reconciliation procedures; these are documented in `analysis/deterministic_reconciliation.json` and `provider_call_audit` artifacts.

VII. CONCLUSION

We preregistered and executed a paired confirmatory study comparing a strong deterministic validator with the same validator augmented by sandboxed emulation on a curated synthetic intent→configuration task bank. Over 720 paired tasks the guarded pipeline produced a negligible aggregate detection gain (0.001388888888888884) with exact McNemar $p = 1.0$ and 95% bootstrap CI [0.0, 0.004166666666666667]. Run accounting shows one repair-stage invocation corresponding to the single guarded-only detection; because `shared_initial_candidate` semantics were enforced, archived per-episode traces permit independent verification of that mapping via `execution/raw_results.jsonl` and `analysis/deterministic_reconciliation.json`. We interpret the result as arising from validator ceiling effects and curated task design; follow-up work should focus on adversarial and production-like task banks, multi-validator ensembles, and higher-fidelity emulation to identify regimes where guarded stages add substantive operational value.

Reproducibility pointers (compact). The canonical execution bundle archived by the run contains: `execution/execution_manifest.json` (execution summary and preregistration SHA-256), `execution/raw_results.jsonl` (input-results SHA-256 `df367ecc40a6f0eaf402129260e11d8450884a4ea5fdb94a602e0d2513aef0e8`), `analysis/deterministic_reconciliation.json` (sha256 `49731a2e52e049ccf5b53c35ddac2351ad75e7e95504dc4a42bcb096d0dec3aa`), `execution/model_configuration.json` (sha256 `a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597ffdd820`), `analysis/paired_contingency_table.csv` (sha256 `efe6e8e7016fa843a8226e911dbde829e943903d719101da8d956b3ff899133c`), and `analysis/condition_summary.csv` (sha256 `394bc690671b731194f9b42b4dcbe28fded6e2ec8cf898b79831225343c22c28`).

The discordant episode is the unique `raw_results.jsonl` entry with `baseline_score = 0` and `guarded_score = 1`; that record contains the `pair_id` and validator/emulation traces that map it to the single repair-stage invocation recorded in `deterministic_reconciliation.json`. These artifacts are archived as the canonical reproducibility bundle referenced by the execution manifest and the preregistration record. (See Disclosure for exact manifest references.)

DISCLOSURE STATEMENT

Disclosure: Autonomous post-lock research run executed under the frozen master prompt supplied to the system. Declared/requested model: "gpt-5-mini" (provider recorded as `openai_responses`). Deterministic

validator: `deterministic_netops_validator_v5`. Orchestration adapter family: `hosted_netops_gvr_v1`. Preregistration id: `f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e` (preregistration SHA-256: `0169919fb8c5aedb2c5a78e031f1ab5c3aeda405cde1ae80fb2f864768595c1b`; recorded in the execution manifest). Initial master-prompt immutable reference: `provenance/master_prompt.txt`, SHA-256 = `1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582b b39b15ba`. Execution summary (artifact-grounded): `completed_episode_count = 1440`, `complete_pair_count = 720`, `model_calls_used = 721` (720 shared initial + 1 repair-stage call); `stage_counts` and `provider-call audit` are archived in `analysis/deterministic_reconciliation.json` (sha256 `49731a2e52e0...`). Human role and autonomy boundary: a human Research Director selected the broad NetOps topic family and constrained the pre-lock master prompt; after the autonomy lock the agentic pipeline autonomously performed literature selection, preregistration, code and prompt generation, experiment execution, analysis, peer review, and manuscript drafting. No manual human editing of technical manuscript content occurred after the autonomy lock. The execution manifest and archived artifacts named above serve as the canonical reproducibility bundle.

REFERENCES

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